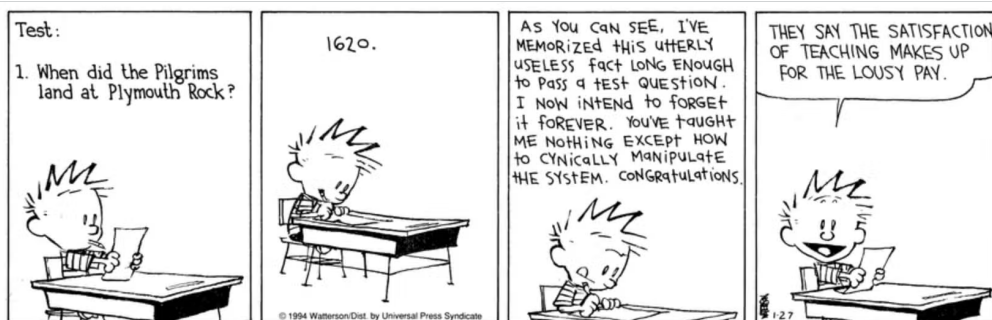
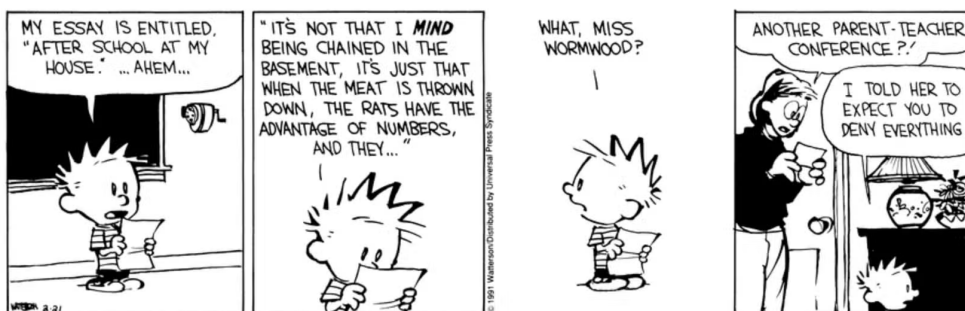


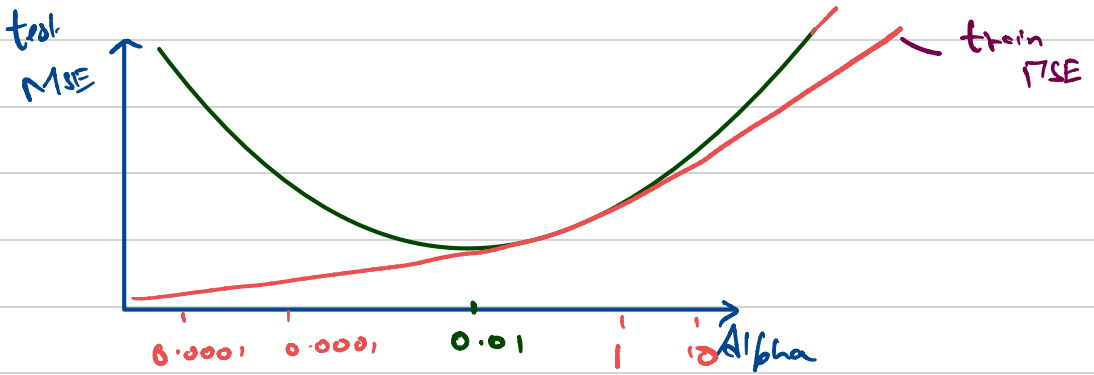
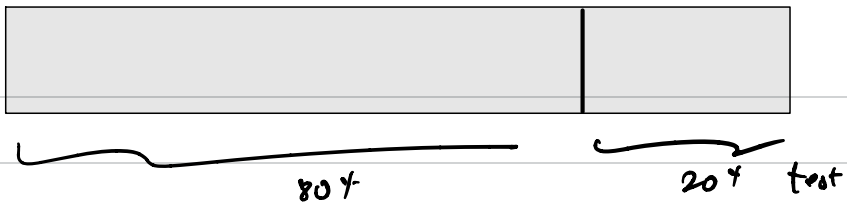


Agenda

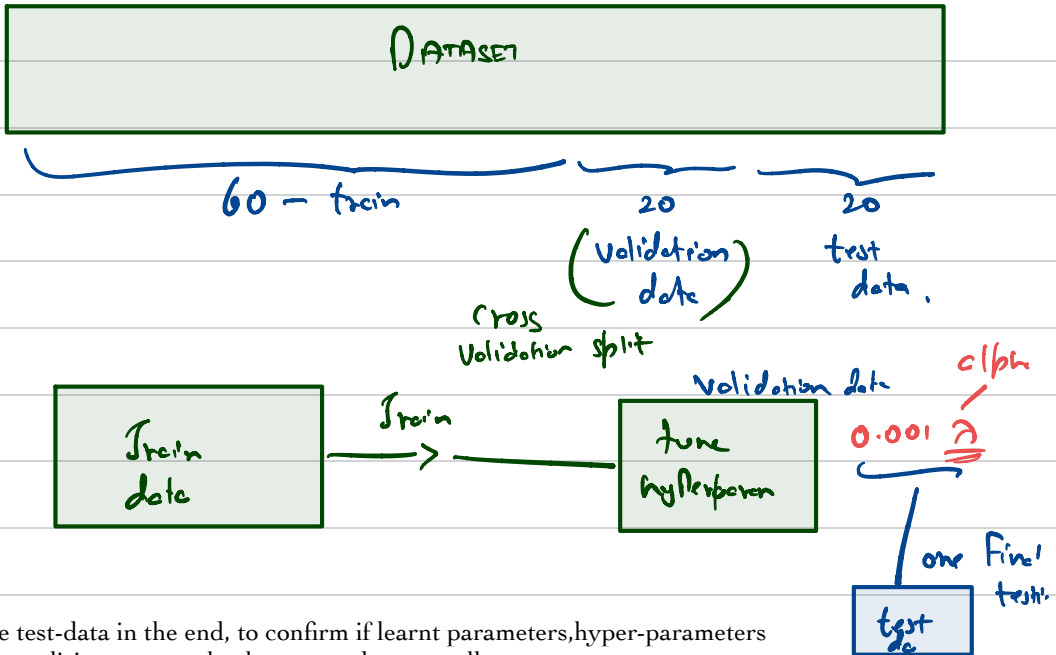
1. Validation/cross-validation Data
2. K-Fold Cross Validation.

GBDT - gradient boosts decision Trees

CS229 - ML
CS231 - CV
CS224 - NLP



Rich man's method.



We use test-data in the end, to confirm if learnt parameters, hyper-parameters Are generalising on completely unseen data as well.

From test-data, we make a final test to confirm if the model is generalising.

$$||\mathbf{w}_E||^2 + \lambda \sum \mathbf{w}^2$$

↓

$$\underbrace{\text{MSE} + 0.01 \times \sum w^2}$$

frein.

✓

0-7 $\Rightarrow$ fest - Validierung des

08 - $t_1 \rightarrow v \quad \gamma$

DRIFT

Poor

Plan's

METHOD

DATA SET

2-Fold Cross Validation

All Data

Training data

Test data

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
	0-20	20-40	40-60	60-80	80-100
Split 1	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 2	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 3	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 4	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 5	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5

Finding Parameters

12 = 5
5-Fold cross validation

Final evaluation

Test data

$$\left(\text{Test MSE} - F_1 + \text{Test MSE} - F_2 + \text{Test MSE} - F_3 + \text{Test MSE} - F_4 + \text{Test MSE} - F_5 \right) \frac{1}{5}$$

All Data

Training data

Test data

$\alpha = 0.01$

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 1	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 2	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 3	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 4	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Split 5	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5

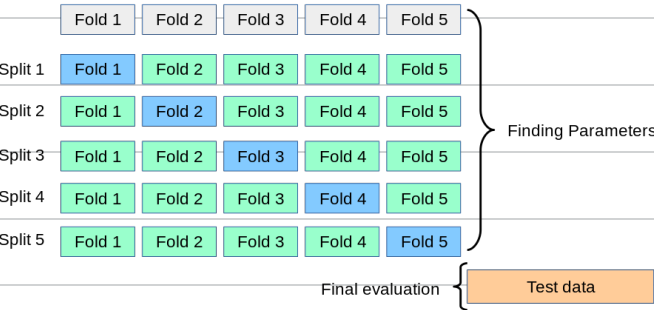
Finding Parameters

Final evaluation

Test data

$$\begin{aligned} L_1 - F_1 - 0.05 \\ L_2 - F_2 - 0.04 \\ L_3 - F_3 - 0.01 \\ L_4 - F_4 - 0.02 \\ L_5 - F_5 - \underline{\underline{0.001}} \end{aligned}$$

≈ 0.0242



<u>$\alpha = 0.01$</u>	<u>$\alpha = 0.1$</u>	<u>$\alpha = 1$</u>	<u>$\alpha = 10$</u>
$L_1 - F_1 - 0.05$	$L_1 - F_1 -$	$L_1 - F_1 -$	$L_1 - F_1 -$
$L_2 - F_2 - 0.04$	$L_2 - F_2 -$	$L_2 - F_2 -$	$L_2 - F_2 -$
$L_3 - F_3 - 0.01$	$L_3 - F_3 -$	$L_3 - F_3 -$	$L_3 - F_3 -$
$L_4 - F_4 - 0.02$	$L_4 - F_4 -$	$L_4 - F_4 -$	$L_4 - F_4 -$
$L_5 - F_5 - \underline{0.001}$	$L_5 - F_5 -$	$L_5 - F_5 -$	$L_5 - F_5 -$
0.0242	0.014	0.001	10

DATASET

60 - train 20 (validation data) 20 test data.

cross validation split

Train data

Train

tune hyperparameters

Validation data

alpha
0.001

one Final test.
test data

$$\alpha = 0.1 \rightarrow L_1$$

$$\alpha = 0.01 \rightarrow L_2$$

$$\alpha = 0.001 \rightarrow L_3$$

Variable

80-20

$$(0.1) L_1 - 0.03$$

$$(0.01) L_2 - 0.01$$

$$(0.001) L_3 - 0.04$$

Rich man

60

Validation data

test-data

$$(0.1) L_1 - 0.03$$

$$(0.01) L_2 - 0.01$$

$$(0.001) L_3 - 0.04$$

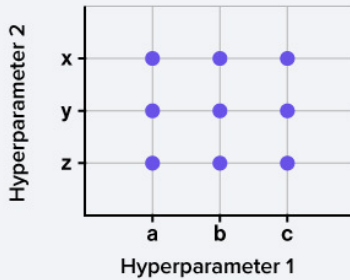
$$0.031$$

$$0.009$$

$$0.041$$

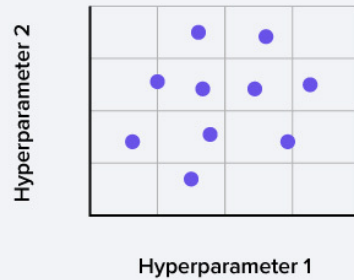
Grid Search

```
Hyperparameter_One = [ a, b, c ]  
Hyperparameter_Two = [ x, y, z ]  
Hyperparameter_X = [ i, j, k ]
```



Random Search

```
Hyperparameter_One = random.num (range)  
Hyperparameter_Two = random.num (range)  
Hyperparameter_X = random.num (range)
```



For p1 in hp_one:

for p2 in hp_two:

for p3 in hp_x:

train_model(p1, p2, p3)

record test_mse for model(p1, p2, p3)

return [p1, p2, p3] for which test_mse was lowest.

Handwritten red notes:
p1 - a → x, y, z
p2 - x, y, z
p3 - x, y, z

